Precision in Algorithms

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Recall and choose your starting point

> Record your starting explanation for Precision in Algorithms before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Expo and Defense

Without opening the old lesson, explain one key idea from Expo and Defense. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Practise with fading help

> Complete the supported Precision in Algorithms trace and record the first step you can explain without help.

Use W2: Predict then observe for From a paper **algorithm** to Python. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Precision in Algorithms explanation against the supplied facts before discussing it.

Use the worked case for From a paper **algorithm** to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Transfer the procedure independently

A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using trays, per_tray and tiles. Predict its **output**. Then change only trays to 0 and predict again. Explain why adding the two quantities is wrong.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Precision in Algorithms target to a trace and a justified conclusion.

Write an **algorithm** for borrowing a classroom book when the book may already be on loan. Include **input**s, a no-copy-available outcome and a check that prevents two borrowers receiving the same copy.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Precision in Algorithms case and show the deciding evidence.

A sandwich **algorithm** says add spread, close it, cut it, without stating materials or directions.

Expose **assumption**s and write a **precise** paper simulation.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A robot starts at square 0. Replace move a bit and stop nearby with instructions that put it exactly on square 3 using steps of one square.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 1 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Precision in Algorithms adds to Transparent Book Recommender and keep the evidence.

Write exact scoring steps so two people rank the same book the same way.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Write an **algorithm** for borrowing a classroom book when the book may already be on loan. Include **input**s, a no-copy-available outcome and a check that prevents two borrowers receiving the same copy.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Connect the paper steps to Python

> Connect the paper steps to Python. Apply the supplied Precision in Algorithms facts and keep your reasoning.

> Write the complete four-line script for 4 packs of 6. Beside each line explain what it does. Which line displays information?

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Predict then observe

> Predict then observe. Apply the supplied Precision in Algorithms facts and keep your reasoning.

Make records for 4 packs of 6, then 5 packs of 6. Predict before running. Record actual **output**, any error and what changed. If working on paper, label both results as traces.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Explain an order error

> Explain an order error. Apply the supplied Precision in Algorithms facts and keep your reasoning.

Move print(total) to the first line of a fresh script. Predict what happens and why. Put the lines back into an order that meets the requirement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Transfer the procedure independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Transfer the procedure independently. Apply the supplied Precision in Algorithms facts and keep your reasoning.

A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using trays, per_tray and tiles. Predict its **output**. Then change only trays to 0 and predict again. Explain why adding the two quantities is wrong.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Keep an honest conclusion

> Keep an honest conclusion. Apply the supplied Precision in Algorithms facts and keep your reasoning.

Explain the difference between a successful run and a correct calculation. State one limit of this **program** and what you would keep so another person can check your work.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Precision in Algorithms

> A fictional book-box counter has three boxes with five books each and two loose books. Explain the labelled step before checking the changed condition.

A fictional book-box counter has three boxes with five books each and two loose books.

Python instructions: boxes=3; total=boxes*5; total=total+2; print(total).

Run the instructions in this order with ordinary integer arithmetic.

AI ACADEMY / AI-2/1

Precision in Algorithms

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read input

boxes is assigned 3.



2. Multiply

total becomes 3 x 5=15.



3. Update total

total becomes its current value 15 plus 2: 17.



4. Print

The program prints 17 once.

CHANGE ONE THING

Set boxes=0, keeping the two loose books.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

If print(total) moves before total=total+2, what gets printed for the original **input**?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Use small predicted traces when translating the Transparent Book Recommender's paper logic into code.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use small predicted traces when translating the Transparent Book Recommender's paper logic into code.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-box counter has three boxes with five books each and two loose books.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The **program** prints 17 once.

> 1. Read **input**

2. Print

3. Multiply

4. Update total

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Print The **program** prints 17 once. **Assignment** replaces a variable's stored value.

In total=total+2, Python first reads the current total on the right, calculates 17, then assigns that result to total on the left. The equal sign is an instruction here, not a claim that 15 and 17 are mathematically equal.

Predict **output** before running.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Set boxes=0, keeping the two loose books. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The printed result is 2.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The printed result is 2.

> 0×5+2=2; there is no rule saying zero boxes means zero loose books.

Repair a missing building block

Review Expo and Defense (ai-1/36). Goal: Evaluate a saved candidate and a manual baseline fairly. Short practice: Write one defensible success statement and one claim the **evidence** cannot support..

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

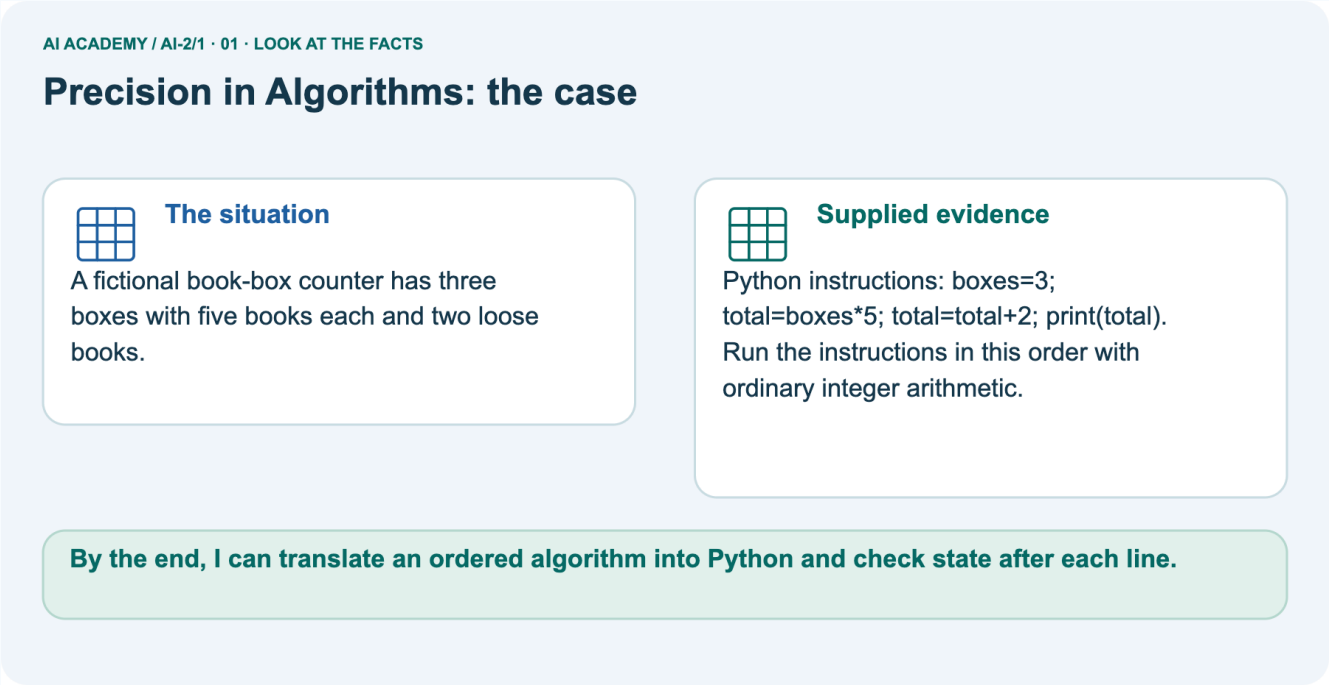
> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Precision in Algorithms picture from its labels, then record a relevant limit.

01. Precision in Algorithms: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

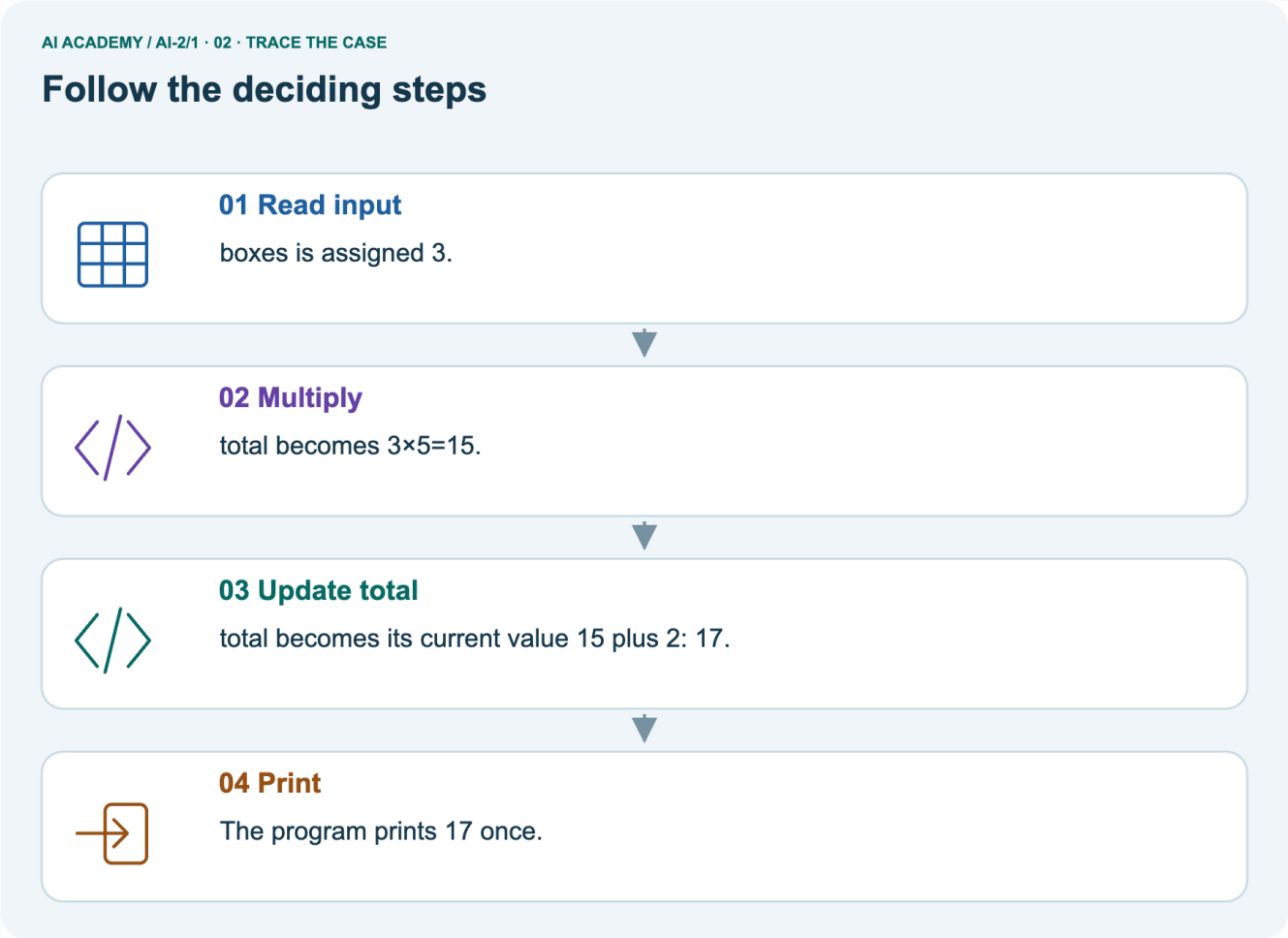
A fictional book-box counter has three boxes with five books each and two loose books. Supplied facts: Python instructions: boxes=3; total=boxes*5; total=total+2; print(total).

Run the instructions in this order with ordinary integer arithmetic.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



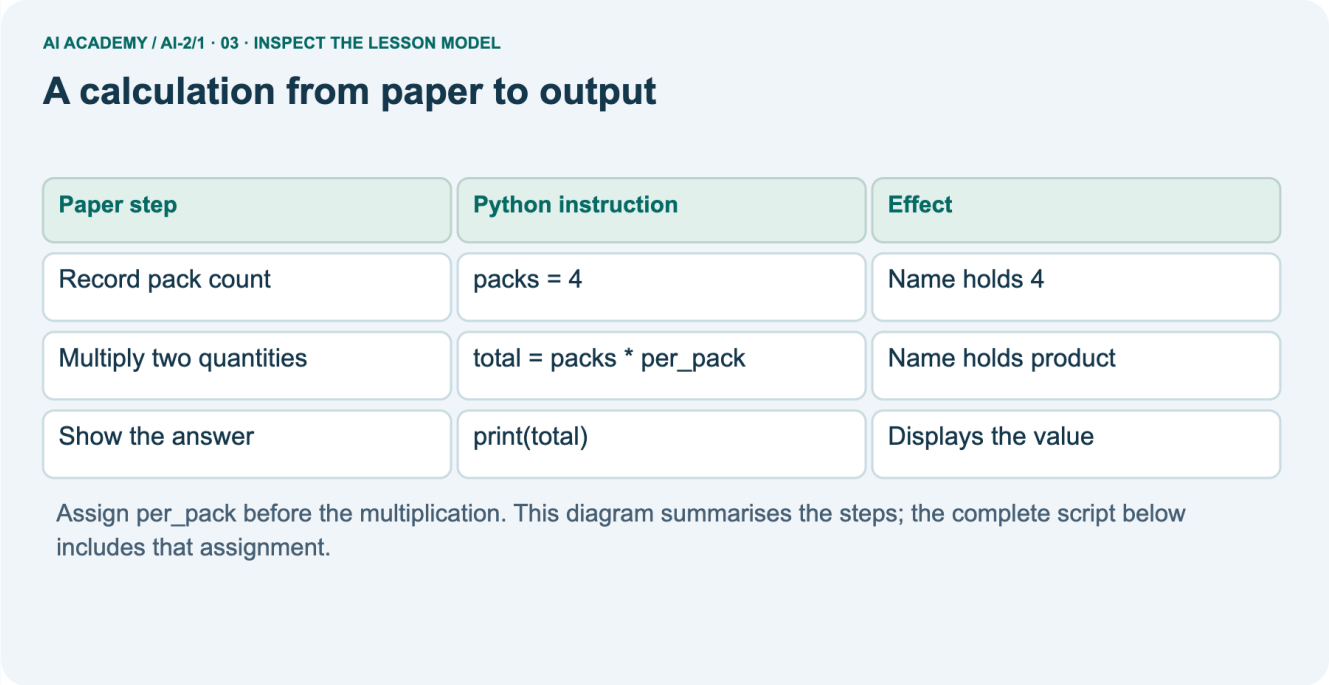
Follow the labelled picture with the matching explanation. Open the vector diagram

Read **input**: boxes is assigned 3. → Multiply: total becomes 3×5=15. → Update total: total becomes its current value 15 plus 2: 17. → Print: The **program** prints 17 once.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A calculation from paper to output



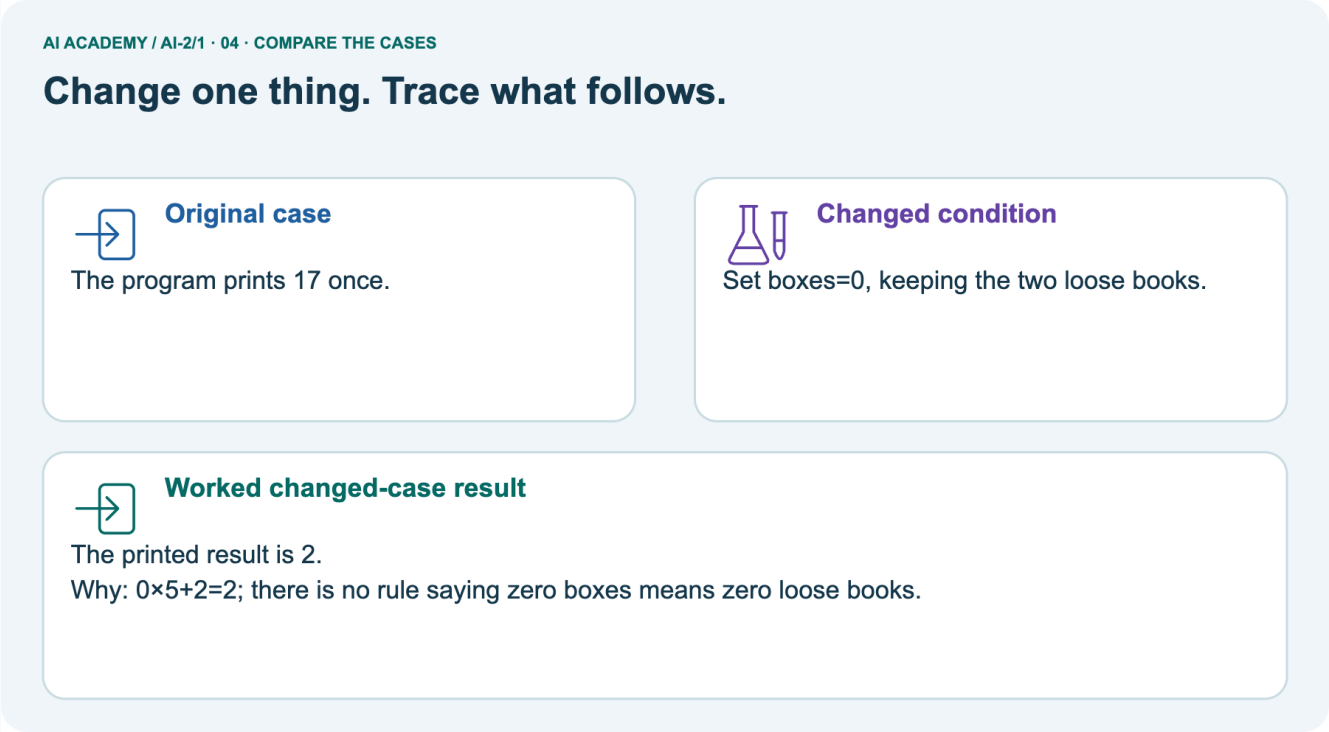
Follow the labelled picture with the matching explanation. Open the vector diagram

A calculation from paper to **output**. Record pack count | packs = 4 | Name holds 4. Multiply two quantities | total = packs * per_pack | Name holds product. Show the answer | print(total) | Displays the value

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



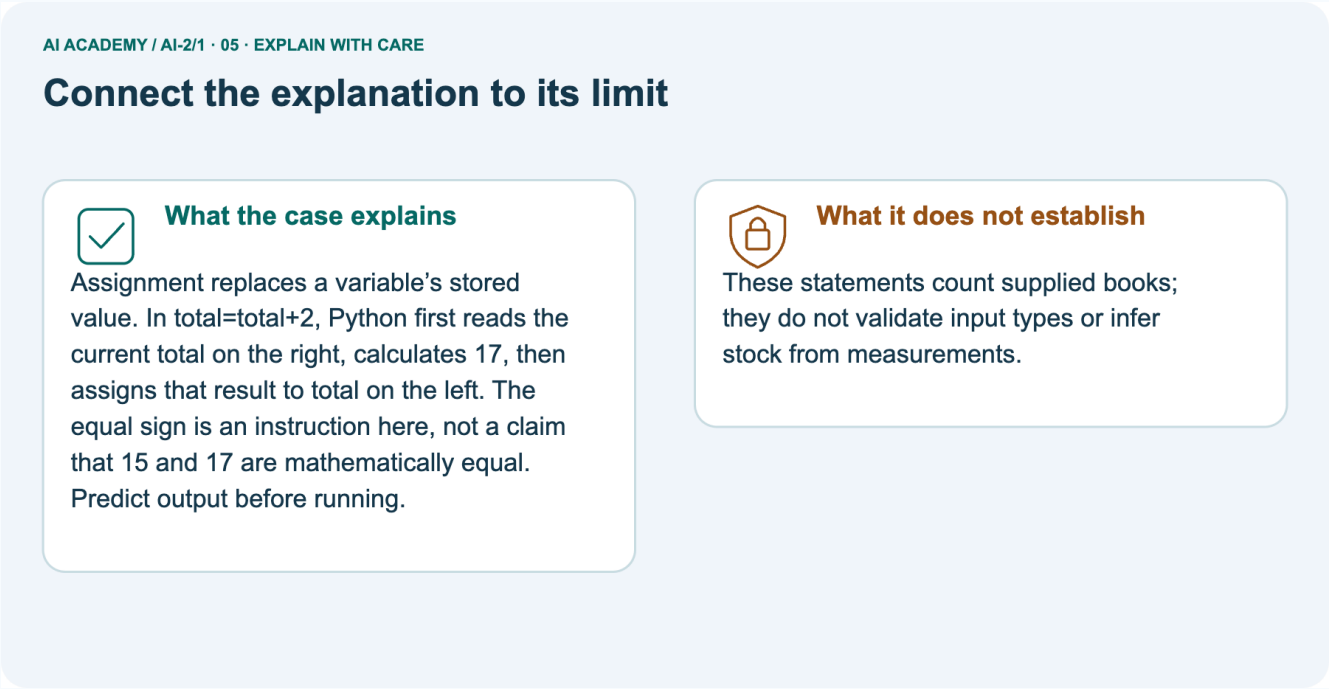
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The **program** prints 17 once. Changed condition: Set boxes=0, keeping the two loose books. Expected result: The printed result is 2. Reason: 0×5+2=2; there is no rule saying zero boxes means zero loose books.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Assignment replaces a variable's stored value. In total=total+2, Python first reads the current total on the right, calculates 17, then assigns that result to total on the left.

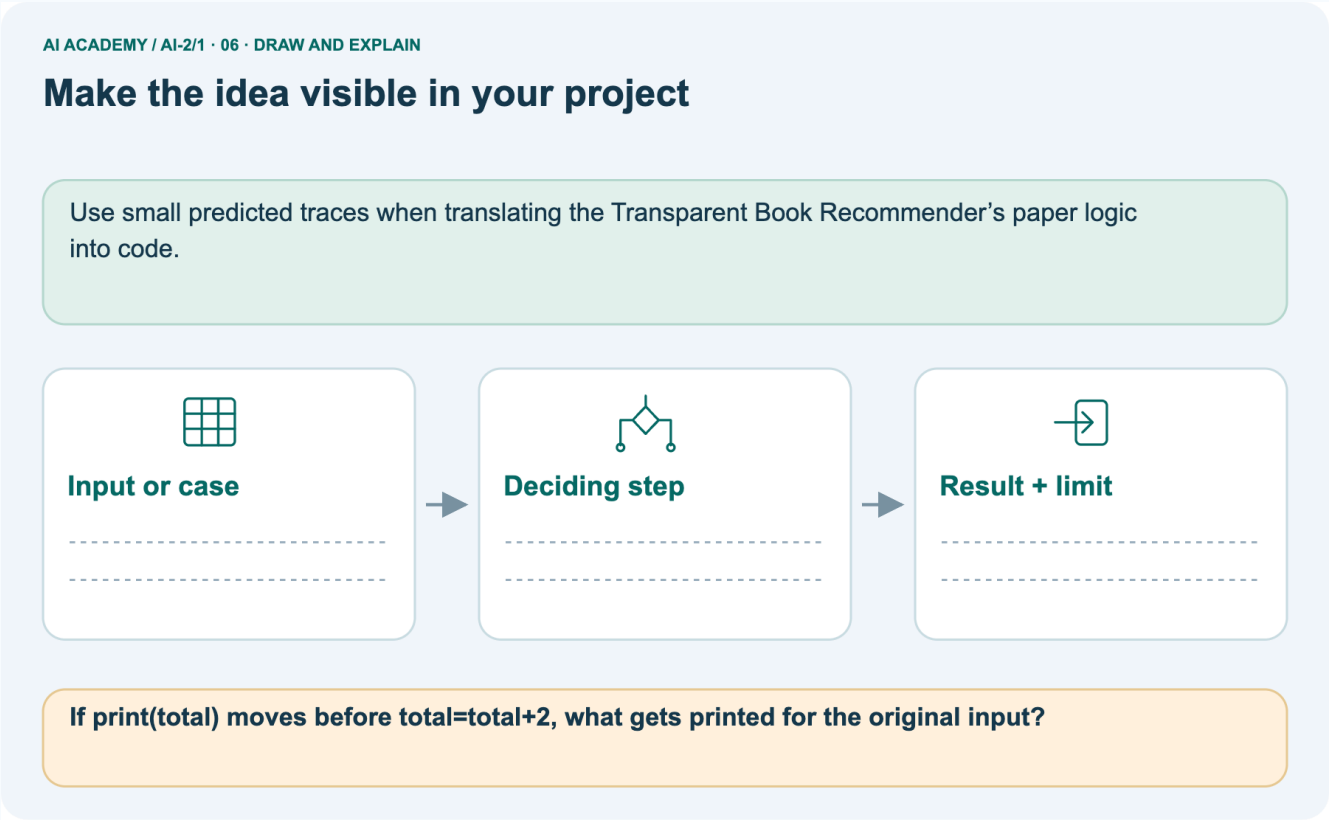
The equal sign is an instruction here, not a claim that 15 and 17 are mathematically equal. Predict **output** before running.

Limit: These statements count supplied books; they do not validate **input** types or infer stock from measurements.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use small predicted traces when translating the Transparent Book Recommender's paper logic into code. Draw your **input**, deciding step, result and limit.

Supported question: If print(total) moves before total=total+2, what gets printed for the original **input**?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.